

Predicting Hotel Booking Cancellations

Improving Revenue Management and Customer Satisfaction

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ABSTRACT

The hospitality industry faces a significant challenge in managing hotel room inventory due to unpredictable booking cancellations. This study aimed not only to predict whether an individual reservation would ultimately be canceled but also to elucidate which factors exert the greatest influence on such outcomes. Drawing upon a large dataset of hotel bookings, we considered a range of attributes—such as lead time, payment methods, booking channels, seasonal trends, and customer demographics—to determine how these elements interact and drive cancellations. By doing so, we sought to answer critical questions: Can we accurately predict cancellations well in advance, and which specific factors should hotel managers focus on to reduce last-minute surprises?

After extensive data cleaning, feature engineering, and exploratory data analysis, we implemented three classification models: Naïve Bayes, Decision Tree, and Logistic Regression. Among these, the Decision Tree model demonstrated the highest predictive performance, achieving an accuracy of around 83.8%. More importantly, we identified a subset of key predictive drivers—most notably deposit type, average daily rate (ADR), and lead time—that significantly affect the likelihood of a booking being canceled. These findings are not purely academic; they have direct implications for hotel revenue management. Hotels can apply these insights to refine their booking policies, adjust pricing strategies proactively, and develop targeted interventions aimed at customers or booking channels that display higher cancellation propensities. Ultimately, our results underscore that a data-driven understanding of cancellation behavior can empower hotels to maintain steadier occupancy rates, optimize staff scheduling, and enhance the guest

experience by minimizing the disruptive effects of unexpected cancellations.

1 Introduction

In the competitive and dynamic environment of the hospitality industry, hotels face continuous pressure to forecast and secure stable occupancy. Unexpected cancellations disrupt these efforts, often forcing reactive measures such as last-minute discounts or risky overbooking strategies. These stop-gap solutions can erode profit margins and strain the guest experience. Against this backdrop, our study addresses a set of important and practical questions: Can machine learning techniques reliably predict which bookings are likely to result in cancellations before the scheduled arrival date? If so, which underlying factors—ranging from booking lead time and deposit requirements to the seasonality of travel—are most influential in shaping that risk? By probing these questions, we aim to bring clarity and predictive power to an inherently uncertain aspect of hotel operations.

Accurately forecasting cancellations holds tremendous value for hotel managers and stakeholders. Predictive insights enable proactive resource allocation and informed decision-making. For instance, if a model identifies that long lead times combined with certain booking channels correlate with higher cancellation probabilities, a manager can adjust inventory strategies, ensure sufficient staffing levels, or introduce targeted incentives to retain bookings well before the critical peak season. In turn, these measures help avoid the negative spiral of last-minute discounts, rushed operational changes, and guest dissatisfaction resulting from overbooking or unfulfilled reservations.

Moreover, from a guest-centric perspective, stable operations supported by accurate cancellation predictions contribute to an improved customer journey. Guests who do choose to stay experience less overcrowding, fewer logistical hiccups, and greater confidence in the hotel's reliability. Equally, the hotel benefits from a reduced administrative burden and can channel efforts into enhancing services, refining marketing campaigns, and nurturing long-term customer loyalty.

In sum, predicting and understanding cancellation behavior is not merely an academic exercise. It empowers hotels to incorporate data-driven insights into their revenue management frameworks, align pricing and promotions with demand patterns, and foster a more seamless and trustworthy booking environment. As a result, both the bottom line of the hotel and the satisfaction of its guests stand to improve, making this research of direct practical importance to the entire hospitality ecosystem.

2 Related Work

The prediction of booking cancellations in the hospitality industry using hotel-specific data is a relatively new area of research. Due to this, there are a limited number of studies published on predicting booking cancellations in the hospitality industry using hotel specific data. Still, there are several studies which have laid the groundwork for understanding the drivers of cancellations and developing predictive models.

One earlier publication is "Forecasting cancellation rates for services booking revenue management using data mining" (2010) by D. Morales and J. Wang [1]. This work explored forecasting cancellations through two-class probability estimation methods. Their key finding was that the importance of predictors varies significantly within the booking horizon. This means an attribute which may be a high predictor of cancellation for a booking with a longer lead time may not be a predictor for a booking with a short lead time. Instead, the high predictor variable for the booking with the short lead time is a different attribute. This temporal dependency highlights the need for dynamic modeling approaches.

Another significant contribution is "Modeling the cancellation behavior of hotel guests" (2018) by M. Falk and M. Vieru [2] which elucidates on factors that contribute to booking cancellations. This analysis looked at individual-level booking data and found a higher probability of cancellation based on booking method, lead time, and country of origin. This work was limited by only analyzing data from a single hotel chain in a specific country and did not include additional attributes which may be of interest to affecting cancellations such as individual characteristics and previous stay history. This study provides some interesting insights into the attributes that may be influential in predicting cancellations to include in our analysis.

The bulk of research on this specific topic is from three co-authors: Nuno Antonio, Ana Almeida, and Luis Nunes. In one of their earlier publications "Using data science to predict hotel booking cancellations," (2016) [3] they showed it is possible to build models for predicting hotel booking cancellations with high accuracy in results. Building on this work, they later published "Predicting hotel bookings cancellation with a machine learning classification model" (2017) [4]. This study employed sophisticated machine learning classification models, such as Random Forests and Gradient Boosting Machines, to predict hotel booking cancellations and analyzed the factors influencing these cancellations. The authors found that variables like lead time, market segment, and previous cancellations significantly impact the likelihood of a booking being canceled. By implementing a machine learning model, the authors showed that predicting booking cancellations is possible from an empirical standpoint.

Building on this foundation, our project seeks to extend these efforts by focusing on model interpretability and the practical implications of predictive factors. Unlike previous studies, we aim to explore the impact of incorporating new features and alternative modeling techniques to improve prediction accuracy and provide deeper insights into customer behavior. Furthermore, our emphasis on

interpretability ensures that the findings are actionable for hotel management.

While other studies have explored cancellation prediction from diverse perspectives—such as the impact of economic indicators or the role of customer sentiment—our project is uniquely positioned. We leverage a dataset from two Portuguese hotels, enabling a detailed analysis tailored to a specific regional and operational context. By emphasizing actionable insights, we aim to bridge the gap between predictive accuracy and practical utility for decision-making in hotel management.

3 Dataset

The dataset used in this analysis is the Hotel Booking Demand Dataset, available on Kaggle at: <https://www.kaggle.com/datasets/mojtaba142/hotel-booking/data>

This dataset originates from two hotels in Portugal: a resort hotel in the Algarve region (H1) and a city hotel in Lisbon (H2). It was collected and published by Nuno Antonio, Ana de Almeida, and Luis Nunes, in collaboration with the hotels.

The data includes 119,390 records (bookings) and 36 attributes. These attributes span a range of data types including nominal, ordinal, binary, interval-scaled, and ratio-scaled. For instance, the field `is_canceled` is a binary data type field where 1 means the reservation was canceled. Additionally, the `lead_time` field, which measures the number of days between booking and reservation, is a ratio-scaled integer that proved to be a significant predictor in our analysis.

The dataset comprises a variety of attributes including:

- **Basic booking details:** hotel type, arrival date, length of stay, number of adults and children
- **Customer and booking characteristics:** `is_canceled`, `lead_time`, `is_repeated_guest`, `previous_cancellations`, `deposit_type`, distribution channels
- **Financials and requests:** `adr` (average daily rate), `total_of_special_requests`, `required_car_parking_spaces`

This dataset offers a comprehensive view of factors influencing hotel booking behavior, making it an excellent foundation for building predictive models to understand and forecast booking cancellations. The diversity of attributes allows for the exploration of complex relationships between booking characteristics, customer preferences, and cancellations.

4 Main Techniques Applied

In alignment with our proposed workflow, data preprocessing was a foundational step in our modeling approach. We began by thoroughly examining the dataset from two Portuguese hotels to ensure data completeness and quality. While the dataset contained minimal missing values, critical fields such as `is_canceled` and `lead_time` were verified to be complete. Irrelevant personal identifiers (e.g., name, email, credit card information) were removed to address privacy concerns and simplify the dataset for modeling. Categorical variables, such as `deposit_type`, `room_type`, and `distribution_channel`, were encoded using either Label Encoding or One-Hot Encoding, enabling machine learning models to process them effectively.

We also created new, domain-informed features to capture potential patterns in the data. For example:

- `total_stay`: Calculated as the sum of weekday and weekend stay lengths, providing a single integrated measure of booking duration.
- `lead_time_log`: A logarithmic transformation applied to `lead_time` to address skewness, stabilize the distribution, and enhance model performance and interpretability.

These feature engineering steps were guided by insights from both prior literature and our Exploratory Data Analysis (EDA). Through EDA, we identified significant relationships, such as the positive correlation between `lead_time` and cancellation rates, as well as the impact of specific distribution channels on booking cancellations.

Although a formal data warehouse or OLAP cube was not implemented, our preprocessing and feature engineering efforts resulted in a well-structured, analysis-ready dataset. By ensuring consistent encoding, handling outliers, and creating informative features, we established a robust “clean data layer” that

supported efficient experimentation with various modeling techniques.

EDA played a critical role in understanding the dataset and preparing it for analysis. Distributions were examined using visualizations such as histograms and boxplots, while correlation matrices provided insights into variable relationships. For instance, `lead_time` demonstrated a clear positive correlation with cancellation rates, and specific distribution channels were strongly associated with cancellations. To address potential multicollinearity, we monitored correlation values and utilized the Variance Inflation Factor (VIF) as needed to detect and mitigate redundant features. These analyses informed feature selection, ensuring the models focused on the most relevant predictors.

To establish performance baselines and evaluate model feasibility, we initially applied three classification algorithms:

1. **Naïve Bayes:** A simple probabilistic model serving as a baseline to gauge whether more sophisticated approaches were warranted.
2. **Decision Tree:** An interpretable model capable of capturing non-linear relationships. This approach aligns with our goal of improving model interpretability for hotel management.
3. **Logistic Regression:** A widely-used linear model that offers interpretability and insight into feature-target relationships. It serves as a benchmark for comparing more complex or non-linear models.

To enhance model generalizability and ensure stable performance estimates, we employed 5-fold cross-validation. This approach verified that performance metrics—such as accuracy, precision, recall, F1-score, and ROC-AUC—were not overly dependent on specific data splits. This rigorous evaluation framework ensured the robustness and stability of our modeling results.

5 Key Results

This analysis aimed to predict hotel booking cancellations and uncover key factors driving cancellation behavior. Using a combination of Decision Trees, Logistic Regression, and Naïve Bayes models, we evaluated model performance and interpretability.

5.1 Model Performance: Logistic Regression

The logistic regression model was evaluated using a combination of hold-out testing and 5-fold cross-validation to assess its performance and generalizability. It provided solid performance as a baseline for predicting hotel booking cancellations:

- **Accuracy:** The model achieved an accuracy of 78.8%, indicating that it correctly classified 78.8% of bookings as either cancellations or non-cancellations.
- **Precision for Cancellations:** 82% of bookings predicted as cancellations were correct.
- **Recall:** 55% of actual cancellations were correctly identified, indicating room for improvement in recall.
- **F1-Score: 66%**

To ensure the model's robustness and stability, 5-fold cross-validation was performed:

- **Mean Cross-Validation Accuracy:** 78.47%
- **Standard Deviation:** 0.34%, reflecting consistent performance across folds.

The logistic regression model provides a solid baseline with good overall performance. The model demonstrated strong precision, correctly identifying 82% of predicted cancellations. However, recall for cancellations was lower, indicating that some actual cancellations were missed. These findings suggest that logistic regression offers a robust and interpretable starting point, though improvements in recall could enhance its predictive value.

5.2 Model Performance: Naïve Bayes

The Naïve Bayes model provided moderate performance for predicting hotel booking cancellations. It served as a simple probabilistic baseline for predicting hotel booking cancellations:

- **Accuracy:** 76.1%
- **Precision for cancellations:** 70%
- **Recall:** 63%
- **F1-Score:** 66%

Cross-validation results demonstrated consistent performance:

- **Mean Cross-Validation Accuracy:** 75.4%
- **Standard Deviation: 0.94%**, reflecting moderate consistency across folds.

The model achieved a reasonable balance between precision and recall for cancellations, making it a simple yet effective probabilistic baseline. While it performed slightly worse than more sophisticated models like Decision Trees and Logistic Regression, it remains a lightweight and interpretable option for quick analysis. These findings suggest that Naïve Bayes is a good starting point but may require enhancements or more advanced methods for improved recall and overall accuracy.

5.3 Model Performance: Decision Tree

The Decision Tree model demonstrated strong performance, outperforming both Logistic Regression and Naïve Bayes for predicting hotel booking cancellations:

- **Accuracy:** The model achieved an accuracy of **85.1%**, correctly classifying 85.1% of bookings as either cancellations or non-cancellations.
- **Precision for Cancellations:** 80% of bookings predicted as cancellations were correct.
- **Recall:** 81% of actual cancellations were correctly identified, indicating strong capability in capturing cancellation cases.
- **F1-Score:** 80%

To ensure the model's robustness, 5-fold cross-validation was performed:

- **Mean Cross-Validation Accuracy:** **83.8%**
- **Standard Deviation:** **0.19%**, reflecting high consistency across folds.

The Decision Tree model demonstrated strong performance across all metrics, with a particularly high recall for cancellations, making it effective at identifying bookings likely to be canceled. Its ability to capture non-linear relationships between features contributed to its superior performance compared to Logistic Regression and Naïve Bayes. These findings suggest that the Decision Tree model is a suitable choice for predicting hotel booking cancellations while maintaining interpretability for hotel management decision-making.

5.4 Significant Predictors

Feature importance analysis revealed several key predictors of booking cancellations, including lead time, deposit type, and average daily rate. These features contributed significantly to the model's ability to differentiate between cancellations and non-cancellations:

- **lead_time:**
 - Longer lead times showed a strong positive correlation with cancellation likelihood. Guests who booked well in advance were more likely to cancel, potentially due to increased uncertainty or changes in plans over time.
 - This feature had the highest importance in the model, contributing **14.3%** to its predictive power.
- **deposit_encoded:**
 - This feature reflects whether a deposit was required for the booking. Non-refundable deposits were strongly associated with a lower likelihood of cancellations, likely because financial commitment reduces the incentive to cancel.
 - It ranked as the second most important feature, contributing **11.0%** to the model's predictive power.
- **adr (Average Daily Rate):**
 - The average daily rate (pricing per room per night) exhibited varying effects on cancellation behavior. Higher rates were associated with increased cancellations for budget-conscious customers, while luxury customers exhibited different patterns.
 - This feature contributed **9.8%** to the model's predictions.
- **country_encoded:**
 - The country of origin was also a significant predictor, contributing **11.9%** to the model. Certain countries showed higher cancellation rates, potentially reflecting cultural differences or varying levels of commitment in booking behaviors.

Lead Time Dominance: The strong impact of lead_time aligns with findings from prior literature, where longer booking horizons typically correlate with higher cancellation probabilities.

Deposit Policies: The significance of deposit_encoded highlights the importance of financial commitment in

mitigating cancellations. Hotels could leverage this insight to design more effective deposit policies.

Customer Segmentation: Attributes like `adr` and `country_encoded` offer opportunities for targeted interventions, such as personalized follow-ups for high-risk segments based on pricing sensitivity or location.

These findings emphasize the importance of dynamic pricing strategies, deposit policy adjustments, and lead-time-based interventions to reduce cancellations and improve revenue management.

5.4 Comparison of Models

The three models—Logistic Regression, Naïve Bayes, and Decision Tree—each had distinct strengths and weaknesses. Logistic Regression provided a solid and interpretable baseline, achieving good precision for cancellations (82%) but lower recall (55%), limiting its ability to capture all cancellations. Naïve Bayes was computationally efficient and had better recall (63%) than Logistic Regression, but its lower precision (70%) resulted in more false positives, and its assumption of feature independence limited performance. The Decision Tree outperformed both models with the highest accuracy (85.1%), recall (81%), and F1-score (80%). It excelled at capturing non-linear relationships and provided intuitive decision paths but is more prone to overfitting and requires careful validation.

Overall, the Decision Tree was the strongest performer, offering the best balance of accuracy and recall for predicting cancellations, making it the most suitable model for this analysis. Logistic Regression remains valuable for its simplicity and interpretability, while Naïve Bayes serves as an efficient baseline for quick comparisons. Future work could explore combining the strengths of these models, such as leveraging Decision Tree performance with the interpretability of Logistic Regression, to optimize predictive power and actionable insights for hotel management.

6 Applications

The insights gleaned from our predictive modeling are not merely theoretical. They have concrete applications that, when integrated into hotel operations, can yield substantial improvements in profitability, guest satisfaction, and overall operational stability. By understanding the probability and drivers of booking

cancellations, hotel management teams can adopt a more proactive, data-informed approach to many aspects of their business.

1. Revenue Management and Dynamic Pricing:

One of the most immediate applications of predictive insights into cancellations lies in the realm of revenue management. Armed with reliable forecasts of how many bookings may fall through, hotels can dynamically adjust room rates to reflect anticipated demand fluctuations. For instance, if the model indicates a high likelihood of cancellations during a particular holiday season or through a specific booking channel, managers can preemptively lower or raise prices to maintain desired occupancy levels. This avoids the costly practice of last-minute discounts or emergency price cuts and instead encourages stable revenue streams by setting prices that reflect the true probability of realized bookings. Additionally, hotels can refine their deposit and refund policies based on predicted cancellation risks, offering non-refundable deposits for higher-risk bookings while maintaining flexibility for trusted clientele. Over time, these policies can shape booking behaviors, reducing uncertainty and stabilizing occupancy rates.

2. Customer Relations and Targeted Incentives:

Beyond pricing strategies, understanding why and when cancellations occur opens opportunities for more personalized and effective customer engagement. If, for example, long lead times combined with certain customer segments or booking channels predict higher cancellation rates, hotels can reach out to these at-risk guests with tailored incentives. These might include special discounts for early arrivals, complimentary amenities, or loyalty program points to encourage guests to honor their reservations. Conversely, the data may reveal that certain guest demographics or repeat customers show a lower propensity to cancel; these guests could be rewarded with consistent perks, reinforcing their sense of trust and loyalty. By aligning incentives with data-driven risk profiles, hotels can shift from a one-size-fits-all approach to more nuanced engagement strategies, ultimately reducing the rate of no-shows and cultivating stronger long-term relationships.

3. Operational Efficiency and Workforce Planning:

Accurate cancellation predictions also translate into more efficient operations. Hotels often struggle with

how many staff members to schedule or how much inventory (e.g., linens, toiletries, in-room amenities) to prepare for a given day. By predicting no-show volumes with greater precision, managers can optimize staffing levels to avoid being under- or over-resourced. This leads to cost savings, as fewer hours are wasted on rooms that remain empty, and also improves staff morale because schedules are less likely to require last-minute changes. Similarly, housekeeping, maintenance, and inventory management teams can plan their work more effectively, reducing unnecessary effort and improving their response to actual guest arrivals.

4. Inventory Control and Resource Allocation:

When hotels have a clearer sense of how cancellations vary across channels, seasons, and lead times, they can allocate their limited inventory more strategically. For example, if a particular channel historically shows a high cancellation rate closer to the check-in date, the hotel may choose to release fewer rooms through that channel in anticipation of last-minute cancellations. Instead, those rooms can be offered to channels with more reliable conversion rates or held for walk-in guests, corporate partners, or loyal customers known to rarely cancel. Over time, data-driven inventory allocation can reduce wasted opportunities, ensuring that as many rooms as possible are booked and actually used.

5. Marketing and Branding Strategies:

Predictive insights into cancellations offer a window into customer behavior that extends beyond the immediate booking horizon. By analyzing factors associated with cancellations—such as time of year, payment methods, customer segments, and booking channels—marketing teams can develop campaigns and outreach efforts that align more closely with guests who follow through on their reservations. This may mean investing more in distribution channels or partners with historically lower cancellation rates, or tailoring promotions to guest segments that have proven more reliable. Over the long run, such alignment helps foster a brand image of reliability, trustworthiness, and a superior guest experience, differentiating the hotel in a competitive market.

6. Scenario Planning and Risk Management:

Cancellations often spike during unforeseen events—economic downturns, sudden travel

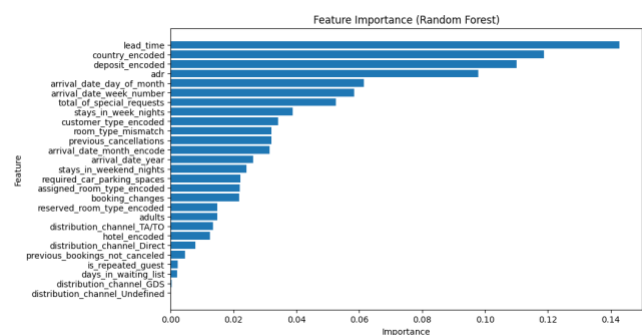
restrictions, or shifts in consumer confidence. Having a robust predictive framework in place enables hotels to conduct scenario planning and stress tests. By simulating how cancellation probabilities might change under different conditions, management can develop contingency plans, ensuring they are better prepared to handle unexpected shocks. This risk mitigation approach fortifies the business against external volatility, maintaining steadier operations even in uncertain times.

7. Continual Improvement through Feedback Loops:

Finally, implementing predictive analytics in cancellation management creates a feedback loop. As hotels act on predictions—adjusting rates, incentivizing certain guests, or allocating inventory differently—new data generated by these actions can be fed back into the model. Over time, the predictive engine becomes increasingly refined, capturing evolving trends, shifting traveler preferences, and the impact of policy changes. This cyclical improvement ensures that the hotel's strategic decisions become more accurate and more aligned with market realities, ultimately leading to sustained improvements in performance and customer satisfaction.

7 Visualizations

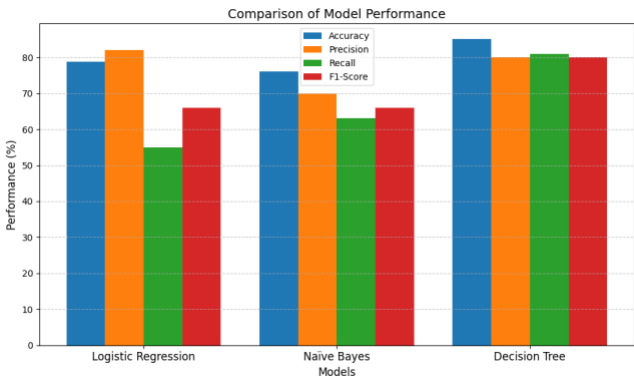
7.1 Feature Importance



This feature importance analysis visualization, generated using a Random Forest model, highlights the most influential predictors of hotel booking cancellations. Lead time is the most significant feature, emphasizing its strong positive correlation with cancellation likelihood. Other key predictors include country_encoded, deposit_encoded, and ADR (average daily rate), reflecting the impact of guest location, deposit policies, and pricing on booking behavior. Attributes such as arrival dates and customer

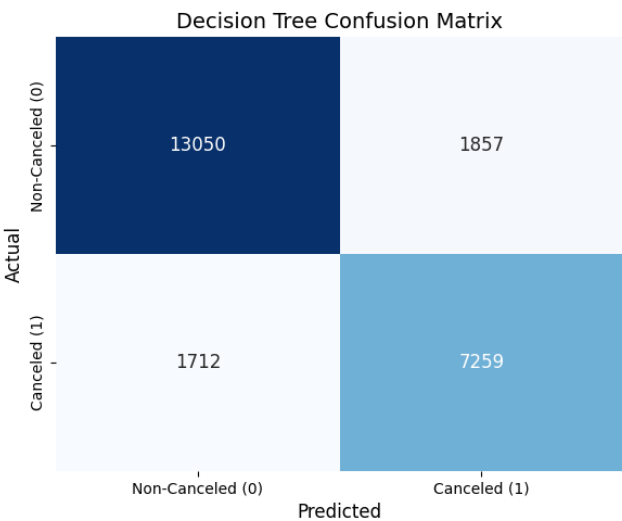
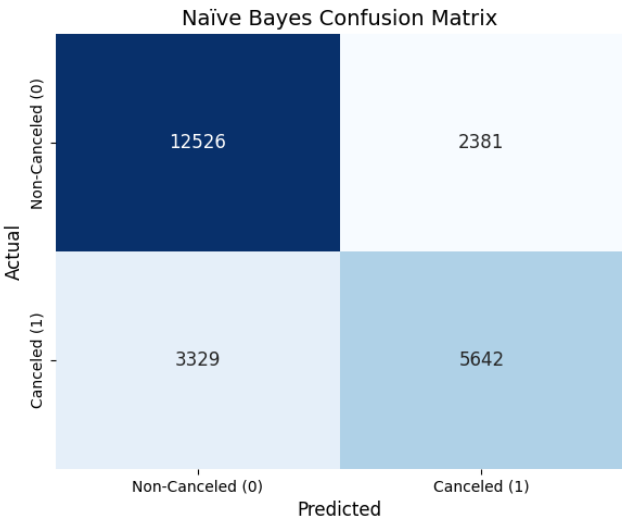
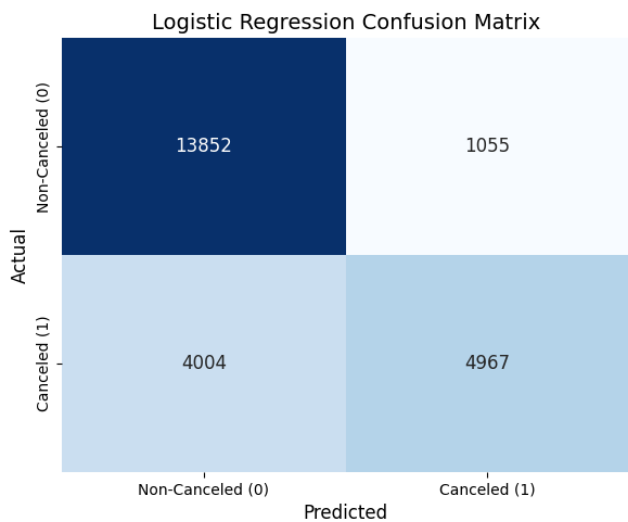
type also contribute meaningfully to predictions, while features like specific distribution channels have lower importance.

7.2 Model Performance Comparison



This bar chart compares the performance of Logistic Regression, Naïve Bayes, and Decision Tree models across four metrics: accuracy, precision, recall, and F1-score. The Decision Tree model outperforms the others, achieving the highest scores in accuracy, recall, and F1-score, indicating its strong overall ability to predict cancellations. Logistic Regression shows a balanced performance with high precision, making it effective at minimizing false positives, though it lags in recall. Naïve Bayes performs reasonably well but has the lowest precision and accuracy among the three, suggesting it is less reliable for this dataset. This comparison highlights the Decision Tree as the most effective model, with Logistic Regression as a viable alternative when precision is prioritized.

7.3 Confusion Matrix Heatmaps



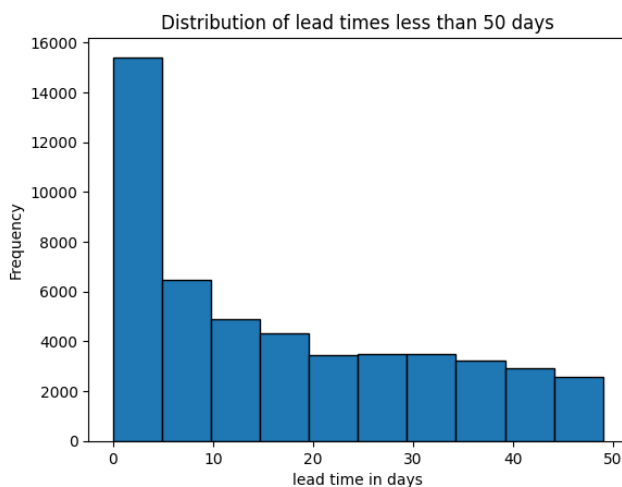
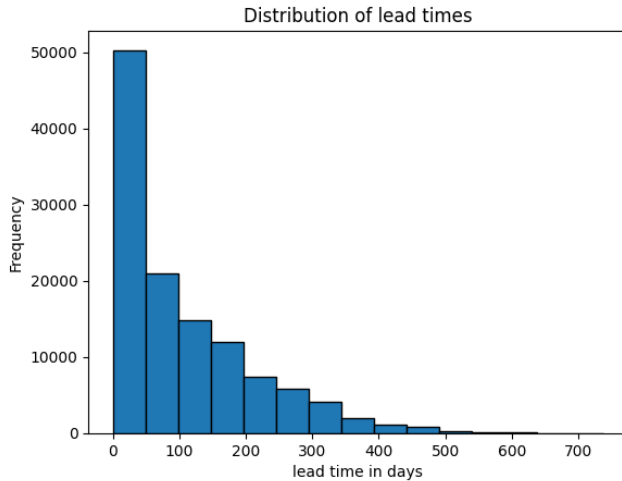
The confusion matrix heatmaps highlight differences in performance among the Logistic Regression, Naïve Bayes, and Decision Tree models. Logistic Regression excels in accurately predicting non-cancellations, with 13,852 true negatives, but struggles with recall for cancellations, missing 4,004 actual cancellations. Naïve Bayes improves on cancellation recall, correctly identifying 5,642 canceled bookings, but this comes at the cost of higher false positives (2,381), reducing its precision compared to Logistic Regression.

The Decision Tree model offers the best balance, achieving the highest number of true positives (7,259 cancellations correctly identified) while maintaining strong performance in predicting non-cancellations (13,050 true negatives). It also minimizes false negatives and false positives more effectively than the other models, making it the most reliable option for

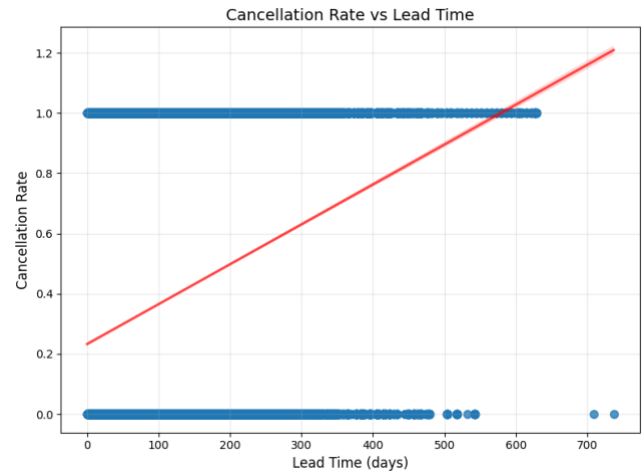
identifying cancellations. These results suggest that while Decision Tree is the most effective overall, Logistic Regression may still be useful for scenarios where reducing false positives is a priority.

7.4 Important Attributes

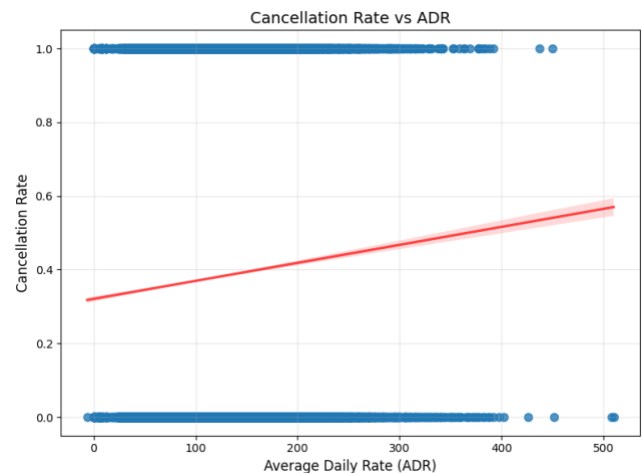
Based on the literature, it was important to evaluate the relationship between cancellations and lead time. Lead time is the number of days that elapsed between the booking and the arrival date. The lead time is left-skewed with a large percentage of reservations made with a short lead time. For instance, 14% of all reservations were made with 5 days or less lead time. On the other hand, only 6% of reservations are made with more than 300 days lead time. This is further illustrated in the histograms below.



7.5 Cancellations by Important Attributes



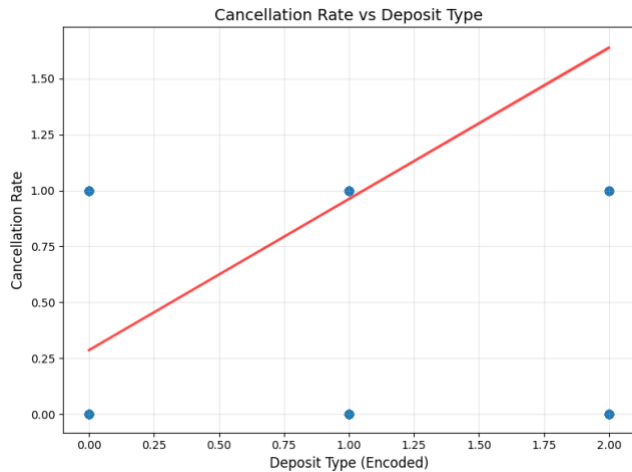
Plotting the cancellation rate versus lead time shows the strong positive correlation between the two attributes. For lead times exceeding 200 days, there is a noticeable rise in cancellation rates, with a significant proportion of bookings being canceled for lead times around 300–500 days.



This scatter plot and trendline illustrates the relationship between Average Daily Rate (ADR) and cancellation rate. The trendline shows a slight positive correlation, indicating that higher room rates are moderately associated with increased likelihood of cancellations. The plot also reveals two prominent clusters: one where cancellations are nearly certain (cancellation rate ≈ 1) and another where cancellations rarely occur (cancellation rate ≈ 0), regardless of ADR. For extreme ADR values (above 400), there are fewer data points, but they suggest varied cancellation behaviors. This relationship highlights the importance of considering pricing strategies when managing

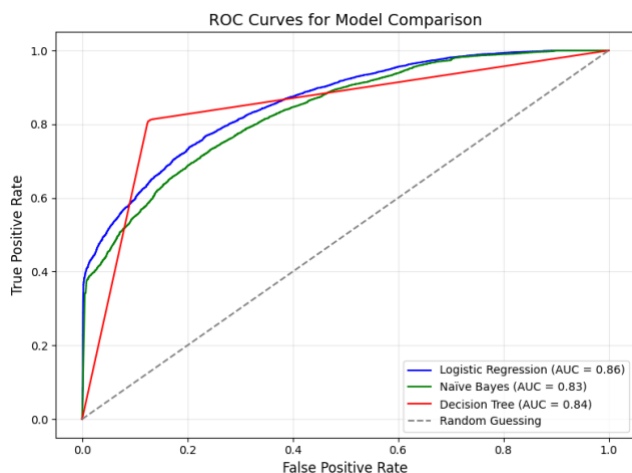
cancellation

risks.



This scatter plot depicts the relationship between deposit type (encoded as No Deposit = 0, Non-Refundable = 1, Refundable = 2) and cancellation rate. The trendline shows a positive relationship, indicating that cancellations are more likely for refundable deposit types compared to non-refundable or no-deposit bookings. Bookings with no deposit required (encoded as 0) exhibit the lowest cancellation rates, while refundable deposits (encoded as 2) are associated with the highest cancellation rates. This highlights the influence of deposit policies on booking behaviors, where financial commitment discourages cancellations. This insight can guide hotels in designing deposit policies to reduce cancellations.

7.6 ROC Curves for Model Comparison



The curves show the trade-off between the true positive rate (sensitivity) and the false positive rate, with the AUC (Area Under Curve) scores included for each

model. This visualization shows the for logistic regression has a good balance between true positive rate (recall) and false positive rate, making it a robust model for general use. The Decision Tree has a strong performance and a steep curve early on, indicating that it captures true positives well with fewer false positives initially.

REFERENCES

- [1] Dolores Romero Morales and Jing Wang, 2010. Forecasting cancellation rates for services booking revenue management using data mining. *European Journal of Operational Research* 202, 554-562. DOI: <https://doi.org/10.1016/j.ejor.2009.06.006>.
- [2] Martin Falk and Markku Vieru, 2018. Modelling the cancellation behaviour of hotel guests. *International Journal of Contemporary Hospitality Management* 30, DOI: <https://doi.org/10.1108/IJCHM-08-2017-0509>.
- [3] N. Antonio, A. de Almeida and L. Nunes, 2016. Using Data Science to Predict Hotel Booking Cancellations. DOI: <https://doi.org/10.4018/978-1-5225-1054-3.ch006>.
- [4] N. Antonio, A. de Almeida and L. Nunes, 2017. Predicting Hotel Bookings Cancellation with a Machine Learning Classification Model. In 16th IEEE International Conference on Machine Learning and Applications (ICMLA), Cancun, Mexico, 1049-1054. DOI: <https://doi.org/10.1109/ICMLA.2017.00-11>.